1. Nyquist Rate | Q1 – Q3

Q1

An audio signal has a highest frequency of 4 kHz. Calculate the minimum sampling rate required.

Given

f_{max} = 4 kHz

Find

Minimum sampling rate (Nyquist rate)

Solution

Nyquist rate = 2 × f_{max}

= 2 × 4 kHz

= 8 kHz

= 8,000 samples/second

✔ Minimum sampling rate = 8 kHz

Q2

The highest frequency in a signal is 8 kHz. What is the Nyquist rate in kHz and samples per second?

Given

f_{max} = 8 kHz

Find

Nyquist rate in kHz and in samples/second

Solution

Nyquist rate = 2 × f_{max}

= 2 × 8 kHz = 16 kHz

1 kHz = 1,000 Hz ⇒ 16 kHz = 16 × 1,000 = 16,000 Hz

1 Hz = 1 sample/second ⇒ 16,000 samples/second

✔ 16 kHz = 16,000 samples/second

Q3

A signal is sampled at 12,000 samples/second. What is the highest frequency it can represent without aliasing, assuming ideal sampling?

Given

f_s = 12,000 samples/second = 12,000 Hz

Find

Highest frequency without aliasing (f_{max})

Solution

$f_s = 2 \times f_{max} \Rightarrow f_{max} = f_s \div 2$

f_{max} = 12,000 ÷ 2

= 6,000 Hz

= 6 kHz

✔ Highest frequency = 6 kHz

Nyquist (Q4) & SNR (Q5 – Q6)

Digital Audio — Practice Problems

Q4

A signal contains frequencies up to 15 kHz. Is a sampling rate of 25 kHz sufficient? Explain briefly.

Given

f_{max} = 15 kHz

f_s = 25 kHz (proposed)

Find

Is 25 kHz sufficient? (Yes / No + reason)

Solution

Required rate = 2 × f_{max}

= 2 × 15 kHz = 30 kHz

Compare: 25 kHz < 30 kHz

Below the Nyquist rate ⇒ aliasing occurs.

✓ No — 25 kHz < the required 30 kHz

Q5

Signal amplitude = 80 V and noise amplitude = 8 V. Calculate SNR in dB.

Given

A_{signal} = 80 V

A_{noise} = 8 V

Find

SNR in dB

Solution

$SNR = 20 \cdot \log_{10}(A_{signal} / A_{noise})$

$= 20 \cdot \log_{10}(80 / 8)$

$= 20 \cdot \log_{10}(10)$

$= 20 \times 1$

✓ SNR = 20 dB

Q6

Signal amplitude = 2 V and noise amplitude = 0.02 V. Calculate SNR in dB.

Given

A_{signal} = 2 V

A_{noise} = 0.02 V

Find

SNR in dB

Solution

$SNR = 20 \cdot \log_{10}(A_{signal} / A_{noise})$

$= 20 \cdot \log_{10}(2 / 0.02)$

$= 20 \cdot \log_{10}(100)$

$= 20 \times 2$

✓ SNR = 40 dB

SNR (Q7 – Q8) & SQNR (Q9)

Q7

Signal amplitude = 50 mV and noise amplitude = 5 mV. Calculate SNR in dB.

Given

A_{signal} = 50 mV

A_{noise} = 5 mV

(same unit ⇒ mV cancels)

Find

SNR in dB

Solution

$SNR = 20 \cdot \log_{10}(A_{signal} / A_{noise})$

$= 20 \cdot \log_{10}(50 / 5)$

$= 20 \cdot \log_{10}(10)$

$= 20 \times 1$

Check: 0.05 V ÷ 0.005 V = 10 ✓

✓ SNR = 20 dB

Q8

The signal amplitude is 10 times the noise amplitude. Find the SNR in dB.

Given

$A_{signal} = 10 \times A_{noise}$

$\Rightarrow A_{signal} / A_{noise} = 10$

Find

SNR in dB

Solution

$SNR = 20 \cdot \log_{10}(A_{signal} / A_{noise})$

$= 20 \cdot \log_{10}(10)$

$= 20 \times 1$

✓ SNR = 20 dB

Q9

Find the maximum SQNR for an 8-bit audio system.

Given

N = 8 bits

(levels = 2^8 = 256)

Find

Maximum SQNR in dB

Solution

$SQNR \approx 6.02 \times N$

$= 6.02 \times 8$

$= 48.16$ dB

✓ SQNR ≈ 48.16 dB

SQNR (Q10 – Q12)

Q10

Calculate the maximum SQNR for a 12-bit system.

Given

N = 12 bits

(levels = 2^{12} = 4,096)

Find

Maximum SQNR in dB

Solution

SQNR ≈ 6.02 × N

= 6.02 × 12

= 72.24 dB

✓ SQNR ≈ 72.24 dB

Q11

Calculate the maximum SQNR for a 16-bit system.

Given

N = 16 bits

(levels = 2^{16} = 65,536)

Find

Maximum SQNR in dB

Solution

SQNR ≈ 6.02 × N

= 6.02 × 16

= 96.32 dB

✓ SQNR ≈ 96.32 dB

Q12

An ideal quantizer has an SQNR of approximately 72.24 dB. Estimate the number of bits per sample.

Given

SQNR = 72.24 dB

Find

N (bits per sample)

Solution

SQNR ≈ 6.02 × N ⇒ N = SQNR ÷ 6.02

N = 72.24 ÷ 6.02

N = 12

Check: 6.02 × 12 = 72.24 ✓

✓ N = 12 bits per sample

Formulas used: Nyquist rate = 2 × f_{max} | SNR = $20 \cdot \log_{10}(A_s/A_n)$ dB | SQNR ≈ 6.02 × N dB

Units: 1 kHz = 1,000 Hz | 1 Hz = 1 sample/second | 1 mV = 0.001 V